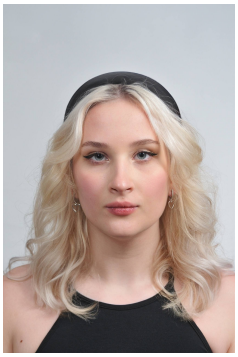
Nataliia Evseeva

Mechanical Design & Product Development

Computer Science Student

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natusikse.github.io/engineering-english/

Profile

Computer Science student at Otto von Guericke University Magdeburg with hands-on experience in mechanical engineering, hardware development, product development, and engineering project coordination. Practical industry experience at Sensata Technologies includes SolidWorks, FEA, prototyping, design validation testing, PLM and BOM management. Strong analytical foundation in mathematics, computer engineering and programming, with independent experience developing a battery monitoring and validation system integrating hardware, software and mechanical design.

Technical Skills

Mechanical Design & CAD: SolidWorks, 3D CAD modeling, mechanical components and assemblies, machine elements, component interfaces, assembly constraints, technical drawings, BOM, BOM verification, GD&T.

Engineering Analysis & Validation: Engineering mechanics, structural analysis, FEA/FEM, component sizing, mechanical measurements, functional and design validation testing, experimental data analysis, root cause analysis.

PLM & Manufacturing Systems: Enovia, Polarion, MES, WARP, engineering change documentation, requirements tracking, technical documentation verification.

Engineering Project Management: Project coordination, change management, cross-functional collaboration, milestone and action-item tracking, supply chain coordination, test scheduling, status dashboards, Asana, Trello, Excel.

Hardware & Embedded Systems: ESP32, voltage/current sensing, temperature monitoring, sensor integration, embedded data acquisition, hardware prototyping and validation.

Materials & Prototyping: Materials science and selection, prototype development, hardware assembly, component integration, troubleshooting, iterative design.

Programming & Automation: Python, C++, C#, Java, SQL, VBA, Power Platform, Power Automate, Power Apps.

Software & Tools: Microsoft Office, Git, GitHub, VS Code, Jupyter Notebook, LaTeX.

Professional Experience

Sensata Technologies Jan 2025 – Jun 2025
Mechanical Design & Product Development Intern Sofia, Bulgaria

- Developed mechanical components and assemblies using SolidWorks and 3D CAD, considering requirements, assembly dependencies, interfaces and manufacturing constraints.
- Performed mechanical calculations and FEA to support component sizing, material selection and engineering decisions.
- Independently sized and selected machine elements including bolted joints, bearings and springs.
- Contributed to prototype development from concept through hardware assembly and integration.
- Executed design validation testing, scheduled tests, tracked results and coordinated improvements with supply chain and production teams.
- Generated technical drawings and BOMs with dimensions, geometric tolerances and component specifications.
- Verified BOM component specifications and documented engineering changes in Enovia, reducing documentation errors by 17%.
- Recorded measurements and test data in MES and WARP to support product verification.
- Investigated prototype issues using root cause analysis and supported design modifications through test cycles.
- Coordinated design changes between engineering, testing and production teams; tracked requirements and modifications in Polarion.
- Tracked milestones and action items, maintained BOM change logs and prepared weekly stakeholder status reports and dashboards.
- Applied programming and automation to repetitive engineering tasks and technical data analysis.

American University in Bulgaria Sep 2025 – Jan 2026
Student Teaching Assistant – Automata Theory Blagoevgrad, Bulgaria

- Developed three Python tools for formal-model analysis and exercise generation for a course with over 40 students, including an AI-assisted generator.
- Held weekly German- and English-language office hours for 10–15 students, supporting algorithmic problem-solving and exam preparation.
- Prepared student progress summaries, tracked coursework completion and supported systematic debugging and evaluation.

Private Tutoring Aug 2024 – Present
Mathematics and English Tutor – Part-Time Remote

- Designed individualized lessons and explained advanced mathematical concepts through structured exercises.
- Adapted teaching methods to student progress and feedback.
- Managed scheduling, progress tracking and parent communication for 50 students, maintaining organized records and weekly summaries.

Education

Otto von Guericke University Magdeburg Apr 2026 – Present
B.Sc. Bilingual Computer Science Magdeburg, Germany
Current GPA: 2.3 (German grading system). Relevant Coursework: Computer Engineering, Algorithms and Data Structures, Database Systems, Automata Theory, Linear Algebra, Recommender Systems, Python for Engineers.

American University in Bulgaria 2024 – 2026
Studies in Computer Science and Mathematics Blagoevgrad, Bulgaria
Studied mathematics, programming and algorithms before transferring to Otto von Guericke University Magdeburg; completed academic credits recognized toward current degree.

Academic Preparation and German Language Studies Jan 2026 – Mar 2026
German language learning and academic preparation Russia
Prepared for university studies in Germany while continuing part-time tutoring in mathematics and English.

Selected Projects

Smart Battery Monitoring & Validation System
Independent Engineering Project ESP32 / Python / SolidWorks

- Designed and developed a low-voltage battery monitoring prototype integrating ESP32 data acquisition, voltage and current sensing, and temperature monitoring.
- Built a Python dashboard to visualize measurements, identify anomalies and generate structured test reports.
- Created SolidWorks CAD models for the enclosure and mounting components, including drawings and a BOM.
- Defined requirements, test procedures and validation criteria, maintaining traceability between requirements and results.
- Tracked milestones, engineering tasks and design changes; maintained documentation using GitHub.
- Performed functional and design validation testing, analyzed data and iteratively improved hardware and software.

Potential Flow Around a Cylinder
Python / Mathematical Modeling / Scientific Visualization

- Implemented and visualized a mathematical model of potential flow around a cylinder, translating equations into a computational flow model.

LLM-Based Support and Knowledge Management System
Python / LLM APIs / RAG / Git

- Built a source-grounded question-answering prototype using a knowledge base of over 100 documents, LLM APIs and retrieval.
- Designed MCP-style tool interfaces and automated document ingestion and answer retrieval.
- Evaluated over 50 AI-model responses for relevance, factual consistency and unsupported claims, refining the retrieval pipeline.

Data Structures and Algorithms in C++
C++ / OOP / Algorithm Design

- Implemented seven data structures including BSTs, AVL trees, splay trees and a min-heap, supporting insertion, search and deletion.
- Implemented Quicksort, Counting Sort and Radix Sort; compared runtime across input sizes and handled pointer-based tree edge cases.

Volunteer Experience

Otto von Guericke University Magdeburg Oct 2026 – Present
Volunteer – Faculty of Computer Science Magdeburg, Germany

- Developed Excel/PowerPoint dashboards and materials for 10–15 research presentations, tracking schedules and milestones.
- Coordinated laboratory equipment and demonstration setups, creating checklists and schedules.

Awards & Achievements

- First Place – National-Level Programming Olympiad, Russia: achieved maximum possible score in a school programming competition.
- Winner – Regional Team Programming Olympiad, Russia: collaborated on algorithmic challenges under time constraints.

Languages

German: B2 (progressing toward C1) | English: C1 | Bulgarian: B2 | Russian: Native